

MATTHEW J. CULLITON

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EDUCATION

Pennsylvania State University College of Engineering

University Park, PA

Major: Computer Science | Graduated: December 2020

Minor: Applied Mathematics and Engineering Leadership Development

SKILLS

Python, Java, Spring Boot, TypeScript, Node.js, React, Angular, LLM APIs (Anthropic Claude, OpenAI), multi-agent orchestration, agentic systems, Kubernetes (AKS), Terraform, Azure Cloud Services, Azure DevOps pipelines, GitHub Actions CI/CD, SQL (Postgres, Vertica), MongoDB, Redis, SQLite / event sourcing, Maven, NPM, SAE Certified, SAE 5.0 Advanced Scrum Master Certified

RELEVANT WORK EXPERIENCE

Aesop — Multi-Agent Orchestration Platform (Independent)

Chicago, IL

Creator & Lead Engineer

2026 – Present

- Architected on Unix principles — small decoupled components, plain durable files, crash-only recovery: event-sourced state layer (SQLite WAL + projections), multi-model driver seam across Anthropic and OpenAI-compatible backends, real-time React/SSE dashboard, MCP server for fleet observability.
- Designed and shipped Aesop (npm-published), a source-available research platform for autonomous software development — flat O(1)-depth dispatch proven across 514 CI-gated PRs: every agent isolated in its own git worktree, no central coordination point; work flows from ranked backlog to CI-verified merge with no human in the loop.
- Ran pre-registered benchmarks across 5 model tiers (equivalence testing); A/B isolation showed context quality — not model tier — drives outcomes (+17pp one-shot fix rate from repro-in-context), and flat dispatch cut costs ~4× vs the cancelled hierarchical design at equal graded quality.
- Engineered reliability and governance: watchdog daemons with heartbeat liveness, kill switch and cost ceilings, fail-closed secret-scan and drift gates, append-only audit logs; every merge gated across Windows/Linux test matrices.

AT&T

Chicago, IL

Software Engineer → Senior Software Engineer, People Analytics

November 2021 – June 2025

- Hired through AT&T's Technology Development Program onto People Analytics, a startup-style HR team building full-stack reporting products on the HR DataHub (Java/Spring Boot, Angular, Azure — millions of rows of SPI); operated as a lead developer from the start and promoted to Senior within the first year.
- Designed HRMAN, a company-wide RBAC service grown from the team's user/role logic, serving tool-access control and real-time reporting-data authorization to teams across AT&T.
- Initiated and led "Canned Reporting": a config-driven reporting platform — report definitions live in the database over one generic frontend and backend; built the POC APIs and ran live demos of report creation, RBAC integration, and compound reports on a Java task-executor pool.
- Shipped the PLE reporting tool for enterprise learning data — 1M+ reports delivered per year — and built stochastic workforce-modeling tools for HR planning.
- Rearchitected backend cross-cutting concerns with Spring AOP — argument resolvers and exception handlers reduced authn/authz to a single annotation; rebuilt low-level result-set processing (HashMap-based) for a 40% memory reduction; introduced Redis caching to the team.
- Grew the team from 5 to 12 developers as its first TDP — recruited through the program, mentored new hires, and wrote the onboarding guide that cut developer ramp-up from 3 days to 1.
- Owned production releases and observability: Azure Application Insights monitoring and tracing saved ~20 developer-hours a month; served 8 months as Senior Scrum Master, lifting team velocity 27%.

The Tannery Row

Chicago, IL

Technology & Operations Lead

December 2018 – Present

- Proposed and launched the company's leather sample-book product line, then scaled it into the full-time program I run today — engineering the swatch-book and leather-panel manufacturing processes and managing staff.

- ➤ Engineered a Google Tag Manager "backend" for a backend-less Squarespace store: dynamic ads, organic listings, and a product-feed API built by treating page visits as tag-firing events through Google's own infrastructure; maintained to this day — online sales +10% YoY.
- ➤ Own SEO end to end: grew organic Google traffic to rival direct traffic for the first time in company history; online sales rose 24% in my first three months back.
- ➤ Built ECM-ai, an accounting tool that gamifies repairing bad order data so invoices import cleanly (a daily manual job became a monthly automated one), and TRTools, an internal RBAC-gated automation toolbox for all employees.

Glimpse (Mobile App)

UI/UX Developer / Co-Founder

2014 – 2018

- ➤ Built and released the social media app "Glimpse" (Swift); won best in Illinois in the Verizon Innovative App Challenge and pitched the app to live audiences and investors — later rebranded and sold after my departure.